A Functional Architecture for Distributed Governance: Mechanism Design, Adversarial Validation, and Computational Evidence for Selective Institutional Distribution

Working paper — v1.11 (July 2026). Revised through successive adversarial and author review cycles, documented in the repository's roadmap. This version consolidates the companion computational program (Offermann 2026b): the deterrence-complementarity rule, the semi-open transition path, the budget-release rule, and machine verification with the human second instance; and the companion's two-layer separation of the macro categorization from the allocation profiles.

Abstract

Debates about the state are usually conducted over its size — more state or less state — rather than its architecture. This paper proposes and defends a functional alternative: decompose state activity into functional layers, distribute the layers where modern coordination technology outperforms institutional monopoly, and keep central the layers where it does not. We present a complete reference architecture (Core v0) for the most contestable layer — distributed allocation within mandate-bound, visible, versioned, and contestable planning scopes: citizens allocate a mandate-recorded share of an existing public budget across verifiable projects, under strict separation of proposing, modeling, executing, evidence-producing, and auditing roles; funds move only through milestone-gated conditional release backed by externally materialized guarantees; and every consequential act leaves a public, layered audit trail. Three contributions distinguish this from the participatory-governance literature it builds on. First, mechanism design: we derive the incentive-compatibility condition for milestone-gated disbursement, a collusion-proofness condition for protocolized fiscalization with k-corroboration, and show formally that weak verification environments must be priced in financial terms — results that convert the architecture's design intuitions into checkable propositions. Second, computational evidence: a 10,000-agent simulation of attention-constrained allocation shows that funding-target closure works as an anti-concentration device but not a quality device, that allocation quality is carried by the informational quality of the project prioritization the passive share follows — the aggregated allocation profiles — rather than by citizen attention, and that participation decay is survivable exactly where that prioritization layer is strong; a pre-registered fourth experiment that models knowledge symmetrically finds that aggregating dispersed citizen signals into that vector outperforms fixed-bandwidth central construction at every tested scale — provided an aggregation institution exists and signals are honest, unbiased, and collected; a pre-registered fifth experiment carries the comparison to what redistribution is for — delivered social value per unit of budget — finding, in the model, that verified delivery adds +43% on identical portfolios, that selection and delivery gains multiply, and that a zero-control regime's officially reported completion overstates real delivery by twenty-nine percentage points; a sixth experiment isolates the incentive channel — visible standards sustain execution quality where opacity lets it decay — and finds that naive reputation-weighted assignment concentrates work faster than it finds ability, evidence for the architecture's concentration-observability machinery; and a pre-registered seventh experiment re-runs the headline against a status-quo baseline calibrated from published audit-institution findings across nine jurisdictions and the European Union, finding, in the model, that the full architecture delivers 2.22× that calibrated baseline per unit of budget at scale and 1.4–1.6× at municipal pilot scale — and that audit at its empirically documented intensity, without reputational memory, deters no diversion at all: it shrinks the reported gap (twenty-nine to nineteen points), never the real one; and a pre-registered eighth experiment replaces the imposed participation side of those arms with adoption trajectories generated by a companion behavioral study, finding the headline unchanged at 2.26× and robust from launch dynamics to adoption-hostile populations. Third, method: the architecture was developed under systematic adversarial review — a structured self-critique of forty attack briefs, each with a paired defense and an accepted resolution under an explicit integrate-or-bound rule that either adds a mechanism through existing objects or records a boundary with a named residual risk, all propagated into the architecture corpus; the final rounds were generated by the method turned on itself — simulated five-profile external reviews of the companion document and of this manuscript, whose unanswerable questions became formal attacks, including the baseline-calibration attack the seventh experiment answers. The result is an architecture whose limitations are part of its specification and whose review record is itself public. We state those limitations plainly: in the closed and tutored operating modes the architecture specifies for pilots, planning agendas are constructed by the implementing authority, and the informational quality of that construction is, by our own simulation, the binding constraint on allocation quality — the designed trajectory is open, socially constructed agenda-setting, whose in-model viability the fourth experiment measures and whose elicitation mechanics remain a gated design problem; fiscal dependence on the incumbent state is measurable but not enforceable; and no empirical pilot has yet been run.

Keywords: distributed governance, participatory budgeting, mechanism design, polycentric governance, public accountability, agent-based simulation, liquid democracy, institutional design.

CONTENTS

1. Introduction

Modern states concentrate three things that do not need to be concentrated together: the authority to decide what public money is spent on, the operational capacity to execute that spending, and the epistemic authority to certify that the spending achieved anything. Where all three sit inside a single institutional hierarchy, accountability depends on the hierarchy auditing itself (Bovens 2007). The predictable consequences — discretionary allocation, self-reported compliance, capture by concentrated interests, and citizen distrust — are documented across every tradition of political economy, from regulatory capture (Stigler 1971; Laffont and Tirole 1991) to the audit society's ritualized compliance (Power 1997) to the distributional coalitions that entrench themselves in stable polities (Olson 1982).

The standard debate responds with quantity: shrink the state or grow it. Both positions treat the state as a single object. This paper argues the tractable question is architectural, not volumetric: *which functional layers of state activity still require central monopoly, and which can now be distributed — with better accountability than the status quo — given that digital coordination has collapsed the transaction costs that once justified hierarchy?* (Coase 1937; Hayek 1945; Williamson 1985) — a knowledge-and-coordination lineage that runs from the calculation debate (Mises 1920) through Hayek's dispersed knowledge to its institutional treatment in Sowell (1980): decisions should sit where the knowledge is, disciplined by feedback rather than articulated authority. Rights guarantees, legitimate force, macro-fiscal stability, and enforceable adjudication plausibly remain central. Information processing, project-level resource allocation, service execution, evidence production, and auditing plausibly do not.

We make the argument concrete rather than programmatic. Core v0 is a complete reference architecture for distributing one bounded layer — the allocation, execution, and verification of a legally mandated share of an existing public budget — developed to the level of named objects, state machines, and decision rules (a corpus of more than one hundred architecture documents, fifty-nine designed hypotheses, and forty adversarial reviews, all public). Citizens receive periodic, non-withdrawable allocation capacity in a civic wallet; projects pass through a parallel-closure lifecycle in which funding, independent fiscalization, evidence commitments, and beneficiary confirmation must all close before execution; the executor never selects or pays its own auditors — removing the agency cost of self-monitored compliance (Jensen and Meckling 1976); money moves only on reviewed milestones with retention and externally materialized guarantees; and every consequential state transition is recorded in a citizen-legible, expert-auditable trail.

The reason to take a design exercise seriously as research is what was done to it. Our contributions are:

1. **A functional distribution principle with an explicit comparative

   discipline.** Critiques of the architecture must be evaluated against how the current institutional model solves the same problem, not against an ideal (Section 2). This blocks the nirvana fallacy in both directions.

1. **Formalization of the core mechanisms** (Section 5). We model

milestone-gated disbursement as a principal-agent game and derive its incentive-compatibility condition; we model bribery of protocol-assigned fiscalizers and derive a collusion-proofness condition under k-corroboration; and we prove two design-relevant comparative statics: weak verification must be compensated with financial terms, and guarantees are counterproductive when detection quality falls below the cost of capital.

1. **Computational evidence** (Section 6). A dependency-free, seeded

   agent-based simulation of 10,000 citizens tests the architecture's behavioral assumptions under rational ignorance, limited discovery attention, and social-proof cascades. The results discipline the design: they support some claims, sharpen others, quantify the leverage concentrated in the scope-construction layer, measure a viable open construction of it, and carry the comparison end to end — from allocation to delivered social value per unit of budget, the criterion redistribution exists to satisfy. The headline result, in the model: verified delivery and social prioritization multiply rather than add, and the full architecture delivers 2.22× a status-quo baseline calibrated from published audit-institution findings (1.4–1.6× at municipal pilot scale), while even that calibrated status quo overstates its real delivery by nineteen percentage points.

1. **Adversarial review as method** (Section 7). The architecture was

   attacked systematically — forty attack briefs grounded in the political-science, economics, and law literatures, each answered by a paired defense and resolved under an explicit integrate-or-bound rule, with every resolution propagated through the corpus and the full review record public by construction. The loop is a structured self-critique, not external validation, and we say so; we propose it, and its terminating rule, as a reusable method for institutional design research.

Section 8 states limitations with the same care as results, because under our method they are results: each is a named, bounded residual risk.

2. The functional distribution principle

We analyze the state as a stack of functional layers rather than a single institution: (a) rights guarantees and legitimate force; (b) binding adjudication; (c) rule-making; (d) macro-fiscal management; (e) macro planning and agenda framing; (f) project prioritization and resource allocation to concrete undertakings; (g) execution and service delivery; (h) evidence production about delivery; and (i) evaluation and accountability. The distribution principle is:

> A layer is a candidate for distribution when three conditions hold:

> its failures under monopoly are information and incentive failures rather

> than coordination-of-force failures; distributed provision can be made

> *more* observable than monopoly provision; and the layer can be bounded so

> that its failure does not cascade into the non-distributable layers.

Layers (a), (b), and (d) fail the first or third condition and remain central in our design. Layers (f) through (i) pass all three, and Core v0 distributes them jointly — jointly, because distributing allocation without distributing verification reproduces PB's delivery gap, and distributing verification without allocation reproduces the audit society.

Layer (e), planning, is the deliberately unresolved case: Core v0 requires planning scopes to be public, versioned, and mandate-bearing, but does not distribute their construction — which is why the architecture's promise is stated with its qualifier built in: what it distributes is allocation *within mandate-bound, visible, versioned, and contestable planning scopes*, never allocation over an unframed agenda. Section 6 shows the qualifier is not a detail: it is the binding constraint of the whole design, and Section 8 elevates its removal to the research program's next object.

Two methodological rules govern everything that follows. The **comparative critique rule** (P001): every objection is evaluated against the current institutional baseline, not an ideal — a difficulty shared by both systems is a general problem of governance, not a refutation of the proposal. The **integrate-or-bound rule** (P007): once the core architecture is complete, a founded objection produces a new mechanism only if the failure mode would defeat a core claim and cannot be controlled through existing objects; otherwise it produces an explicit boundary, a visible residual risk, and a limitation statement. The first rule disciplines critics; the second disciplines the designers — a bias toward few, simple, general rules over proliferating specific machinery in the spirit of Epstein (1995).

3. Related work

Polycentric governance. Ostrom's demonstration that common-pool resources can be governed by nested, self-organized regimes without central monopoly (Ostrom 1990) is the closest intellectual ancestor: her design principles — bounded scope, monitoring by accountable monitors, graduated sanctions, cheap conflict resolution — reappear here as software-enforced objects. The difference is scope and instrument: we target budgeted state functions rather than natural-resource commons, and encode the regime in a platform whose rule changes are themselves versioned, auditable objects.

Participatory budgeting. Porto Alegre-style PB delegates a share of a municipal budget to citizen assemblies (Wampler 2007; Baiocchi and Ganuza 2017). Empirically, PB improves some fiscal outcomes but suffers from engagement decay, capture by organized minorities, and weak links between allocation and verified delivery (Peixoto and Fox 2016). Core v0 differs on exactly those margins: allocation is continuous and individual rather than assembly-based; delivery is bound to allocation through evidential contracts and conditional disbursement; and the architecture treats low participation as a design input (Section 6) rather than a failure to be exhorted away — collective action does not sustain itself at scale (Olson 1965). The civic wallet itself generalizes the voucher mechanism (Friedman 1962) — citizen-directed public money — from choosing among service providers to allocating among verifiable projects, adding the verification lifecycle that vouchers never carried.

Fiscal federalism and epistemic democracy. The formal ancestors of the functional distribution principle are the decentralization literature — Oates's (1972) theorem on when decentralized provision dominates, Tiebout (1956) on preference revelation through jurisdiction choice, and Besley and Coate (2003) on centralized versus decentralized provision under political economy — with one deliberate difference: our layers are functional rather than territorial, so what is distributed is a stage of the spending process (allocation, execution, verification) rather than a level of government. On the epistemic side, the aggregation results of Section 6 belong to the lineage of Condorcet's (1785) jury theorem and its modern failure conditions (Austen-Smith and Banks 1996) — a debt we make explicit, because the theorem's failure regime (correlated, strategic, or biased signals) is exactly what the seventh experiment stress-tests — and the design conversation with Landemore's (2020) open democracy and Fung and Wright's (2003) empowered participatory governance is direct: those programs distribute deliberation and empowerment; this one distributes allocation, execution, and verification with a mechanism-design and audit-trail core they do not attempt.

Liquid democracy. Transitive or scoped delegation promises flexibility between direct and representative participation, at the cost of concentration (Blum and Zuber 2016; Kahng, Mackenzie and Procaccia 2018). Core v0 adopts scoped, revocable, non-compensated delegation with mandatory concentration visibility, and — following Michels' (1911) warning rather than dismissing it — treats delegate concentration as a monitored risk with stress thresholds, not a solved problem.

Digital and blockchain governance. The DAO literature demonstrated both the feasibility of rule-encoded collective resource allocation and its characteristic failure: plutocratic token voting and governance capture (De Filippi and Wright 2018). Core v0 is deliberately not a blockchain design — identity is verified rather than pseudonymous, and the sovereign state remains the source of funds and law — but it imports the lesson that meta-governance is the highest-leverage attack surface (Section 8).

Mechanism design and audit. Our formal models are elementary applications of moral hazard under imperfect observation (Holmström 1979) and collusion in supervision hierarchies (Laffont and Tirole 1991), with Goodhart's law (Goodhart 1975; Campbell 1976) as the standing warning against metric gaming. The closest existing mechanism design for citizen allocation of public funds is quadratic (plural) funding (Buterin, Hitzig and Weyl 2019), which prices concentration through matching-fund curvature; Core v0's funding-target closure rule pursues the same anti-concentration goal by truncation rather than pricing, and our simulation results (Section 6, Finding 1) delimit what truncation can and cannot achieve. On the audit side, Olken's (2007) field experiment on Indonesian road projects is the canonical empirical anchor for the detection probabilities our Propositions 1–2 take as parameters — and its finding that top-down audit outperformed grassroots monitoring for procurement fraud is a caution this architecture absorbs by making professional fiscalization, not crowd observation, the release-gating layer. The Brazilian audit-lottery literature (Ferraz and Finan 2008) supplies the complementary mechanism evidence — disclosure of audit findings changes political outcomes, and audit exposure reduces subsequent corruption — and its underlying CGU data enter the seventh experiment's calibrated baseline directly. The contribution here is not technical depth but specificity: the models' parameters map one-to-one onto named architectural objects, so every proposition is an implementable dial.

What is new. To our knowledge, no prior work combines (i) a functional decomposition of state activity into distributable and non-distributable layers, (ii) a complete object-level architecture for the allocation layer, (iii) formal incentive analysis of that architecture's specific mechanisms, (iv) behavioral simulation of its citizen-facing assumptions — including, to our knowledge, the first symmetric-knowledge comparison in which open construction of allocation priorities from aggregated citizen signals is measured against bandwidth-constrained central construction, and wins at every tested scale — and (v) a documented adversarial-review method with an explicit stopping rule. And the substantive result is itself new: (vi) an end-to-end institutional comparison on the criterion redistribution exists to satisfy — delivered social value per unit of budget — decomposing selection from delivery on matched portfolios, showing the two layers multiply, and introducing the visibility gap (officially reported minus real delivery) as a measurable accountability deficit of the status quo — a comparison then (vii) recalibrated against a baseline parameterized exclusively from the published findings of supreme audit institutions across nine jurisdictions and the European Union, under a pre-registered headline-withdrawal condition. Participatory-budgeting evaluations measure participation and allocation; audit studies measure leakage after the fact; we know of none that measures, within one framework, how much delivered value an allocation institution produces from the same resources.

4. The Core v0 architecture

We summarize the reference architecture at the level needed for the analysis; the full object model, state machines, and citizen-interface layers are specified in the public corpus.

Funding. An implementing authority migrates a mandated share of an existing budget into individual civic wallets: periodic, non-withdrawable, public-purpose allocation capacity, equal per citizen by default. Every active planning scope carries an *Allocation Mandate* record naming the statute or instrument that authorized the migration, its legal rank, the organ to which allocations are imputed, and the allocation formula — the platform records external authorization; it does not manufacture it, and binding-mode operation is gated on an enabling norm of sufficient rank being recorded, with consultative or tutored operation as the disclosed lawful default otherwise. The citizen's allocation act carries the full electoral bargain: vote-like immunity and vote-like secrecy. Individual allocations are pseudonymous at the public layer and reconcile cryptographically against public per-scope totals — every peso traceable as money, no citizen traceable as an allocator, and no receipt or exportable proof of any individual allocation exists, even voluntarily, so that a patron who demands proof can never get one (the secret ballot's own defense, applied to the wallet). A *Fiscal Commitment Profile* per scope makes the migrated percentage, indexation, and delivery latency public and versioned, so fiscal strangulation by the incumbent treasury is measurable and attributable rather than silent. Essential services with continuity obligations are protected by non-assignable floors outside citizen-by-citizen popularity.

Projects and roles. Financeable projects declare a value thesis with verifiable claims, affected parties, risks and anti-values, a phase and milestone plan, and an *evidential contract*: what must be proven, by what class of qualified producer, with what method, for which formal effect. Six roles are structurally separated — proposer, modeler/designer, executor, fiscalizer, evidence producer, custodian — with related-party relationships declared on a severity-classified graph. The load-bearing rule is that the executor never chooses or pays its own fiscalizers or evidence producers: control work is financed from a separated control budget and assigned by protocol.

Parallel closure and conditional disbursement. A published project gathers funding commitments, fiscalizer assignments, evidence commitments, and beneficiary confirmations concurrently; execution becomes possible only when all conditions required by its proportional *threshold policy* close. Committed funds are custodied, not transferred: release happens per milestone, against reviewed fulfillment evidence, with retention, blocker checks, and guarantees materialized by external custodians before any release. A *Duty-of-Care Anchor* names, before disbursement, the solvent legal person answerable to third parties for physical harm.

Attention infrastructure. Citizens act through a layered interface: discovery with user-controlled, reason-visible ordering; compact project cards; and progressively deeper audit surfaces down to the full trail. Non-attending citizens are served by configurable automatic allocation profiles — or a sensible default profile when none is set — and by scoped, revocable delegation with concentration visibility. The architecture does not assume attentive citizens; it assumes mostly inattentive ones and routes their weight through inspectable intermediation (Lupia and McCubbins 1998). This is a design answer to the citizen-competence objection in its sharpest contemporary form (Brennan 2016): rather than restricting anyone's standing, the architecture makes the intermediation that inattention produces visible, revocable, and auditable.

Transition. Deployment proceeds through operating modes — closed, tutored, semi-open, open — in which a public authority may retain admissibility review, but every material tutored decision, and every tutored silence past its deadline, becomes a public governance-resolution object. Incumbent-resistance indicators (scope share opened, rejection and timeout rates, operator privilege) make symbolic adoption distinguishable from real transfer.

5. Formal analysis

We state the three models and their results; proofs are one-step algebra and appear in the companion note (`research/formal-models.md`). All agents are risk-neutral; budgets are normalized to 1. The deterrence structure throughout is Becker's (1968): a violation is deterred when the detection probability times the stake at risk exceeds its gain — our contribution is mapping each term of that inequality onto a named, configurable architectural object.

5.1 Milestone-gated disbursement

An executor chooses to deliver a milestone at private cost $c \in (0, 1)$ or to divert. The mechanism releases an advance a, holds the remainder for reviewed acceptance, recovers a fraction r of the advance on confirmed non-delivery, seizes a posted guarantee γ (carrying cost κ per unit), and imposes a reputational continuation loss R. Review confirms diversion with probability p — the joint quality of evidence standards, fiscalizer independence, and corroboration.

Proposition 1 (incentive compatibility). Delivery is optimal iff

> $c \leq p \cdot [\ (1 - a(1 - r)) + \gamma + R\ ].$

Delivery must be cheaper than the detection probability times the total stake at risk. Every disbursement-gaming test in the architecture is a term in this inequality.

Proposition 2 (weak verification must be priced). The minimal guarantee sustaining delivery for all $c \leq \bar{c}$ is $\gamma\backslash($p$) = max\{0,$ $\bar{c}$/p $- (1 -$ a$(1 -$ r$))$ $-$ R$\}$, *decreasing and convex in* p*. Where verification is weak — thin fiscalization markets, poor evidence quality — the mechanism must compensate with smaller advances, more recoverability, larger guarantees, or higher-reputation executors. A single global guarantee percentage cannot be optimal across heterogeneous verification environments.

The deterrence terms of this condition are complements, not substitutes. A pre-registered ablation on the companion experiments (Offermann 2026b) measured the consequence: at the designed operating point the inequality holds with slack, so removing any single term costs almost nothing — and a deployment negotiated one defensible concession at a time can cross the threshold invisibly, ending below the status quo it replaced (0.87× verified value with selection quality intact) while looking fully instrumented. The corpus therefore requires every scope to publish its deterrence-inequality margin in its operating configuration, recomputed on every term change, with term reductions classified as material rule changes (docs/111). The same slack, kept intact, buys an unexpected dividend: it absorbs verification- instrument error — in the companion panel of five real model families, even a machine verifier passing ~20% of judgment-requiring fraud produced leakage indistinguishable from pure-human verification, because the cascade removes the attempts upstream — provided the adversary is not colluding across the verification layers (Offermann 2026b, docs/113).

Proposition 3 (participation-deterrence trade-off). Raising γ relaxes incentive compatibility at rate p but lowers honest executors' payoff at rate κ; under a designer objective weighing one unit of incentive slack equally against one unit of honest-executor payoff, a guarantee increase is net-beneficial only if $p > \kappa$ (other weightings shift the threshold, not the trade-off's structure). Where detection quality is below the local cost of capital, piling on guarantees excludes honest low-margin executors without deterring fraud — the formal content of the architecture's proportionality discipline.

5.2 Collusion-proof fiscalization

A non-delivered milestone is worth G to the executor if fraudulently approved. Release requires approval by k protocol-assigned fiscalizers, each carrying forfeitable stake W (future protocol fees plus role reputation) and facing post-approval discovery probability q. Because assignment is protocolized and repeat pairings are visible, executor and fiscalizer are strangers: a bribe offer is itself reported with probability δ, costing the executor penalty P_e.

Proposition 4 (collusion-proofness). Approval fraud is unsustainable if

> $k \cdot q \cdot W \geq G - (\delta / (1 - \delta)) \cdot P\_e$,

and in particular whenever $kqW \geq G$. Three corollaries carry design weight. *Corroboration substitutes for reputation capital*: the required stake per fiscalizer falls linearly in k, so redundant review is exactly what makes shallow-reputation fiscalizer pools workable, at linear control cost — which is what proportional threshold policies are for. *Repeat relationships are the attack surface*: the friction term exists only while relational contracting is prevented, which is why repeat-pairing visibility is load-bearing (we hold the report probability δ exogenous to k; endogenizing it — more approached reviewers, more chances of a report — only strengthens the condition). *Thin markets attack both models at once*: a monopolist fiscalizer that cannot credibly be excluded loses its forfeitable stake ($W \to 0$) while also degrading p in Proposition 1 — the two conditions identify the same environments as fragile, for the same reason.

5.3 Attention-constrained allocation

Citizens allocate small individual budgets; the pivotal return to evaluating projects is negligible, so rational ignorance is the equilibrium (Downs 1957), and the design question is where the *inattentive* majority's weight flows: to salience amplified by social proof (Bikhchandani, Hirshleifer and Welch 1992; Salganik, Dodds and Watts 2006), or to the architecture's own default layer, whose routing follows the distributed project prioritization — the aggregated allocation profiles — not a central plan. The model yields three testable predictions — caps tame cascades (P1), defaults anchor quality (P2), decay degrades gracefully only with defaults (P3) — evaluated next.

6. Computational evidence

We simulate 10,000 citizens over 24 monthly cycles allocating across a standing pool of 40 projects with quality θ, salience s (measured corr(θ, s) ≈ 0.24), prioritization need-weights $w = \lambda\theta + (1 - \lambda)u$ with mixing weight $\lambda \in \{0.4, 0.8\}$ — measured corr(θ, w) ≈ 0.55 and ≈ 0.97 respectively — and 3× scarcity (only a minority of projects can complete). Evaluators (2–10%) fund the best quality they sample; salience followers see a six-slot discovery surface ranked by salience amplified by funding progress; default followers' budget fills projects in planning- priority order. The funding-target closure rule is toggleable. Twenty seeded runs per condition; the code is dependency-free and deterministic (`scripts/simulation/allocation-sim.mjs`; full tables in `research/simulation-results.md`).

Finding 1: funding caps are an anti-concentration device, not a quality device. With closure ON, concentration falls (funding Gini 0.732 vs 0.759), the most salient 5% of projects absorb less (16.8% vs 19.6% of all funding), and 15% more projects complete (25.3% vs 21.9%). But quality selection is unchanged (sel(θ) ≈ 0.39 vs 0.41): the truncated surplus spills to the next most *visible* project, not the next *best* one. The architecture's claim for the closure rule should be — and in the corpus now is — bounded accordingly.

Finding 2: the default anchor, not citizen attention, carries allocation quality. A default-anchored mix with a near-perfectly informed planner (r ≈ 0.97) reaches sel(θ) ≈ 0.71 — roughly 1.6–2× the salience-driven configurations (≈ 0.35–0.43) — while quintupling citizen attention (α from 2% to

10%) moves quality selection by at most ≈ 0.08 in salience-driven regimes and essentially nothing in default-anchored ones. Degrading the vector's informational quality from near-perfect to moderate ($r \approx 0.97 \rightarrow 0.55$) costs ≈ 0.29 of quality selection — far more than any feasible attention gain recovers. Two qualifications keep this finding honest. The dominance of defaults is partly by construction — the default rule is a deterministic θ-correlated allocator holding most of the budget — so the informative content is the *conditioning*: how much the vector's informational quality determines the anchor's value, and how little citizen attention substitutes for it. And a sensitivity panel (varying evaluator sample size and social-proof strength) shows the regime ordering is robust except under very strong social proof, where the regimes converge within noise because strong amplification also propagates the evaluators' quality signal; magnitudes are parameter-dependent and uncalibrated. What survives all variations is the ordering and the dominance of the prioritization's informational quality — which quantifies the leverage concentrated in whatever constructs the project prioritization the passive share follows. That prioritization has two layers the companion's engine at first fused and has since separated (Offermann 2026b): the macro categorization — this corpus's Planning Scope, which frames eligibility and carries no budget weights — and the aggregated allocation profiles that route budget within it; distributing the categorization matters most precisely where the inattentive majority routes by value-orthogonal rules. Two architectural facts scope that statement correctly, and an earlier draft's phrasing ("the architecture's central vulnerability... whoever constructs planning scopes holds the allocation's quality in their hands") lacked both. First, the default layer is pluggable, not mandatorily central: the corpus's civic autopilot gives each citizen manual allocation, delegation, published allocation profiles, a personal automatic rule, or the system default — an onboarding citizen must explicitly select or acknowledge a base profile, and only the share of citizens who never engage necessarily follows the system default, which itself operates under a recorded Allocation Mandate. Second, centralized construction of scope weights is a property of the closed and tutored transition modes, not of the architecture: operating modes are country-configured states, and the designed trajectory is toward open construction (Finding 4 measures its in-model viability). What the numbers establish is therefore a conditional: the informational quality of the **project prioritization the passive share follows** — the aggregated allocation profiles, not a macro planning vector — is the binding constraint on allocation quality, whoever or whatever supplies it; a captured or ignorant supplier is the failure mode, a well-informed or well-aggregated one is the asset, and randomizing that prioritization to escape capture would buy neutrality at the price of near-random quality for the passive share. The binding constraint is thus the quality of that prioritization — and because the architecture's designed trajectory distributes its construction (open mode) and keeps it visible, versioned, and pluggable, the constraint is met by distribution, not by a central agenda. This is distinct from, and should not be conflated with, the narrower agenda-setting point of Section 8, which concerns only who frames eligibility. Two things E1–E3 cannot say, and an earlier draft over-read them to say: the prioritization's origin is unspecified (r is a property of the vector, not of a state office), and the modeled crowd carries social proof but no knowledge — so these experiments compare attention against weight quality, not central against distributed knowledge. Finding 4 was designed, after author review, to make that comparison properly.

Finding 3: what buffers participation decay is the anchor's level, not where the leavers' weight flows — our own prediction failed here. We predicted that decaying active evaluation (10% to 2% over 24 cycles) would degrade allocation gracefully only if the released share flowed to defaults rather than salience. Exploiting common seeds across conditions, the paired analysis rejects this: the destination's effect on overall selection is null at both anchor strengths (mean paired differences 0.001 [−0.031, 0.033] and 0.021 [−0.028, 0.071]), and the only interval excluding zero is small and opposite-signed (at a strong anchor, the salience destination preserves late-cycle alignment slightly better — plausibly because amplified social proof also propagates evaluator seeding). What governs robustness to decay is the structural share of the default layer itself, Finding 2's variable. Engagement decay — the documented fate of civic-tech participation (Hirschman 1970; Peixoto and Fox 2016) — is a buffered risk here, but the buffer is the institutional layer's size, and within-cycle quality alignment still erodes in later cycles under all conditions, so decay is bought, not free.

Finding 4: aggregated dispersed signals outperform fixed-bandwidth central construction of the weight vector at every tested scale — but only through an aggregation institution. A fourth, pre-registered experiment (design and predictions committed before any run; `research/e4-institutional-knowledge-design.md`) models knowledge symmetrically instead of endowing it: a planner with fixed bandwidth (thirty deep inspections; its correlation with latent quality is now measured output, collapsing 0.81 → 0.37 → 0.17 as the project pool grows 40 → 200 → 1000) against thirty percent of citizens holding four individually poor local signals each (noise 0.35). Five regimes share the identical world and signals and differ only in the aggregation institution. The pre-registered scale-crossover prediction failed in the informative direction: open construction of the weight vector — a plain average of citizen signals per project — beats pure central construction at *every* scale, including the smallest, where the planner inspects three-quarters of the world (sel(θ) 0.76 vs 0.62 at N = 40; 0.73 vs 0.04 at N = 1000). Twelve thousand noisy signals average into a near-perfect vector; thirty good inspections cannot compete, and fixed central bandwidth decays toward randomness as the world grows — a Condorcet (1785) aggregation logic, and we treat it as such: the jury theorem's known failure conditions (Austen-Smith and Banks 1996) define exactly the boundary the seventh experiment tests. Three qualifications carry the finding's honest weight. First, the same dispersed knowledge *without* an aggregation institution is wasted: the uncoordinated regime funds 0.6–15% of projects and selects poorly — the result vindicates aggregation mechanisms, not the absence of mechanism, and Core v0's default-plus-closure layer is exactly such a mechanism. Second, aggregation defeats noise, not bias: signals are unbiased by construction, and a common-mode bias uncorrelated with quality (misinformation, expressive allocation) would not average out — only endogenous social-proof contamination was tested, and proved largely benign because visible funding progress is itself quality-correlated in a well-anchored system. Third, elicitation is non-strategic by assumption; in deployment, signal reporting becomes a gaming and clientelism surface, and the mechanics of open scope construction remain a gated design problem. Within those bounds, the finding reverses the reading an earlier draft invited: the binding variable is not who holds the pen but how much dispersed information the scope-construction institution ingests.

The simulation also disciplines rhetoric — in both directions. Nothing in E1–E3 supports describing Core v0 allocation as "the wisdom of crowds": the honest description is *inspectable intermediation with a citizen-correctable default*, which the results show is both realistic and better than the salience-driven alternative that unstructured platforms converge to. And nothing in E1–E3 licenses the opposite reading — that central planning knowledge beats distributed knowledge — because they never modeled distributed knowledge; when E4 does, aggregation wins wherever its named preconditions hold. Both discourses lose their slogan; the design keeps its numbers.

Finding 5: delivered value, not allocation, is where the architecture earns its keep — and selection and delivery multiply. A fifth, pre-registered experiment (`research/e5-value-delivery-design.md`) adds the execution stage the first four omitted: executors with hidden types whose diversion decision follows Proposition 1's incentive condition, under an opaque delivery regime — a zero-control lower bound (low detection, unprotected advances, no recovery, no reputational memory), with parameters inside the empirically documented leakage band: 87% in Uganda's capitation grants (Reinikka and Svensson 2004), 24% in Indonesian roads (Olken 2007) — versus the verified regime built from Propositions 1-4 (milestone gating, retention, recovery, evidence-layer detection, and a reputational stake: a visible confirmed-diversion record that costs future selection by funders). Crossing delivery with the two E4 selection regimes yields a 2×2 whose main effects are two plain questions. Same projects, different control layer: the verified regime delivers +43% on identical portfolios (paired ΔV = 0.168 [0.143, 0.193]), and the zero-control regime's official completion overstates its real delivery by twenty-nine percentage points. Same control layer, different projects: social prioritization delivers +53-54% under either regime. The interaction is positive and significant (+0.085 [0.053, 0.117]): the layers multiply, and the full architecture delivers, in the model, 2.19× the zero-control lower bound per unit of budget (0.859 vs 0.393), robustly across direct-participation rates from 3% (the participatory-budgeting floor) to 40% (voting-scale) — a bound recalibrated against the audit-anchored status quo in Finding 7. Two pre-registered predictions failed honestly. The expected dominance of delivery over selection did not hold at this scale — central selection at two hundred projects is near-random (E4), inflating the selection margin — so the robust claim is the compound, not the ranking. And the expected reputational cleansing never fired under the strong verification parameters, for the best possible reason: the incentive condition holds for every executor, so no one diverts and there is no one to exclude — deterrence pre-empts punishment, Becker's enforcement working ex ante. A labeled post-hoc sensitivity with weakened verification shows the second line of defense: partial deterrence, active detection, and an executor pool that measurably improves as funders stop selecting confirmed diverters, whose records are visible (opportunist share 0.28 → 0.21 over twenty-four cycles) — pool exit by lost preference, not by any platform exclusion power; reputation informs choices, it never excludes. A companion sweep of default discovery categories shows each carries a large, measurable distributional signature — near-me concentrates 71% of budget in the densest quintile, rural inverts it — so the default is a visible, configurable distributional policy lever, not an inherent planner bias.

Finding 6: visibility sustains the standard; naive reputation markets concentrate faster than they select. A sixth pre-registered experiment (`research/e6-reputational-competition-design.md`) isolates the incentive channel from deterrence entirely — a career-concerns setting in Holmström's (1999) sense: an all-honest executor pool with adjustable effort and no possibility of diversion (the model prices no explicit effort cost; cost-minimization is encoded behaviorally as the opaque regime's decay rule). In the opaque regime, effort collapses toward cost-minimization (0.49 → 0.24 over twenty-four cycles) — not through malice, but because effort has no return and no visible standard exists to imitate. Making verified quality visible sustains effort near its starting level, and the visible-competitive regime delivers +11% over the opaque baseline — a pure incentive gain with zero diversion in the model. Two pre-registered predictions failed informatively. Visibility alone carries most of the effect: the mirror precedes the market (the behavioral rule ties imitation to visibility, so part of this is by construction — but the construction encodes the claim that opacity erodes professional norms by removing the standard itself). And naive reputation-weighted assignment concentrates work dramatically (assignment Gini 0.84 versus 0.34) while tracking true ability only weakly — early-luck lock-in, the cumulative-advantage dynamic of information cascades reappearing in the executor market. The design lesson runs in both directions: verified visibility is where the quality incentive lives, and any strong reputation weighting — human or algorithmic — needs the concentration observability, entrant floors, and opportunity-normalized reputation the corpus prescribes. In Core v0, reputation informs funders' choices rather than automatic assignment, with concentration visible by construction — and it never excludes: no protocol rule bars a funder from choosing any admissible actor on reputational grounds.

Finding 7: the headline survives the incumbent's own auditors. The manuscript-review round's sharpest attack held that the zero-control baseline is a caricature — real administrations run audit institutions, retentions, bonds, and inspection — and the answer was a seventh pre-registered experiment (`research/e7-calibrated-baseline-design.md`) with a committed withdrawal condition: if the headline collapsed against a fair baseline, it would be withdrawn, not requalified. The calibrated status-quo arm is parameterized exclusively from published audit-institution findings — detection from Chile's comptroller works studies, retention from documented payment-state practice, recovery from Mexico's ASF series, leakage anchors category-matched to construction (Olken 2007; the multi-country evidence base spans GAO, NAO, the European Court of Auditors, Brazil's TCU and CGU, and the comptrollers of Chile, Peru and Colombia; Ferraz and Finan 2008) — with the planner's inspection bandwidth scaled to scope and coordinated signal bias swept as the Condorcet failure regime. The withdrawal condition was not triggered: in the model, the full architecture delivers 2.22× [2.10, 2.35] the calibrated status quo per unit of budget at scale, and 1.4–1.6× at municipal pilot scale (10-40 projects), where central selection with full coverage is competitive and the case rests on delivery and metering. The experiment's central finding strengthens the thesis more than the surviving multiplier does: audit at its empirically documented intensity, without reputational memory, deters no diversion at all — the calibrated regime's incentive threshold lies below every opportunist's cost, so its leak equals the zero-control regime's, and what real-world detection buys is a smaller lie (the visibility gap falls from twenty-nine to nineteen points), never more delivery. The calibrated arm's leak lands inside the audit-documented band (24-48% in works), so the model's leak mechanics, fed audit parameters, reproduce the empirically observed reality. And the bias sweep bounds the open-construction claim honestly: distributed selection degrades near-linearly with coordinated signal capture and crosses below a competent full-coverage municipal planner only at roughly a thirty percent coordinated share — it degrades, never collapses, and remains superior everywhere below ten percent.

A pre-registered eighth experiment (E8, `research/e8-behavioral-participation-design.md`) then replaced the participation side of these arms — the default share and informed share the architecture arms had imposed — with adoption trajectories generated by a companion behavioral study: a Core v0-conformant agent-based model of awareness, registration, participation modes, and trusted microdelegation, calibrated with LLM-elicited synthetic priors (replication package: the distributed-governance-experiments repository). The headline survives: 2.26 [2.23, 2.30] at scale under the calibrated population and 2.15–2.9× across three populations and all scales, including a launch trajectory that begins near zero participation — which costs 1.7% of the ratio, because the default layer anchors the thin early cycles by construction. The behavioral study also independently reproduces the informed-share assumption these experiments had imposed: 0.309 emergent against the 0.30 assumed.

7. Adversarial review as method

The architecture was developed under a documented adversarial loop: **attack** (a brief stating a failure mode, its location in the corpus, a stress scenario, and literature anchors) → **paired defense** (an objective evaluation classifying the attack as founded, partially founded, or a difference of judgment, with line-anchored citations into the corpus) → **resolution** (an accepted document that either integrates a mechanism or bounds the risk) → **propagation** (the resolution's constraints threaded through every affected architecture document). The loop ran four rounds, all now fully resolved and propagated. The first round: eighteen attacks on the architecture's mechanisms (metric gaming, fiscalizer capture, disbursement gaming, collusion, related-party control, complexity, incumbent resistance, among others). The second: fifteen deliberately deeper attacks on the political and behavioral foundations (democratic mandate, agenda-setting, fiscal dependence, thin markets, meta-governance vacuum, rational ignorance, cascades, clientelism, polarization, intertemporal myopia, the problem of many hands (Thompson 1980)). The third round emerged from the method turned outward: a simulated five-profile external review of this paper's companion document generated reviewer questions the corpus could not answer with existing anchors, and the standing rule converted the two serious ones into formal attacks — the legal characterization of the citizen allocation act, and the administrative capacity cost of tutored operation — both since resolved and propagated. The fourth round turned the same instrument on this manuscript itself: five simulated reviewer profiles (academic, public law, systems architecture, public-sector practice, educated general reader) attacked the published v1.6, and their five unanswerable objections became formal attacks, each now resolved — the zero-control baseline as a calibration strawman (answered by the seventh experiment and a binding reporting rule), the reserve of law over the allocation competence (an enabling-norm record gating binding mode), reputational exclusion as an unprocessed sanction (reclassified: the design holds no exclusion power to sanction with), allocation traceability against preference secrecy (resolved as citizen allocation secrecy with public money), and the adoption paradox (an adoption layer under an explicit thesis boundary). The method's honesty requirement applies to itself: several resolutions answer their attacks with an explicit "bounded, not solved," and the full review record is public.

The loop terminates by the integrate-or-bound rule (P007). Its output discipline is what distinguishes it from ordinary threat modeling: every bounded attack must leave three artifacts — an explicit boundary sentence ("Core v0 does not require X"), a visible residual risk, and a one-sentence limitation statement. The limitations section below is therefore not a gesture of humility; it is the accumulated, adversarially generated output of the method. Of the forty attacks, none was dismissed; nine of the second round's fifteen were classified founded outright and the other six partially founded, all five of the manuscript round were classified founded at least in part, and the corpus's answer to several is an honest "bounded, not solved."

We used the loop with a single design team plus AI assistance — which is why we call it structured self-critique rather than validation; a self-administered adversary, however disciplined, cannot substitute for independent attack. Its obvious next application is with genuinely independent reviewers, which we identify below as the first item of future work.

8. Limitations

Stated per the method's own rule — each is a recorded boundary with a named residual risk.

Constructing the eligibility frame is centralized in the transition modes. In the closed and tutored operating modes Core v0 specifies for pilots, the implementing authority constructs planning scopes; the architecture makes that construction public, versioned, mandate-bearing, and contestable through visibility, but in those modes it does not distribute it. Constructing the scope means defining the frame — which purposes, which budget share, which protected floors, which admissibility rules — not designing or ranking projects: project creation and prioritization remain distributed even in tutored mode, so this residual agenda power is the power to decide what *may* be funded, never what *is* funded. It is important not to misread our own simulation here. What that simulation shows dominating every other quality margin is the informational quality of the **project prioritization** — the aggregated allocation profiles the funded share follows — and that prioritization is distributed by design, even in tutored mode; the result is therefore an argument *for* distributing construction, not evidence that a central agenda governs delivery. The residual centralized power is the narrower one: constructing the eligibility frame is itself the second face of power (Bachrach and Baratz 1962; Schattschneider 1960; Lukes 1974) — the power to keep something off the menu — which the architecture answers, in these modes, by making the frame public, versioned, mandate-bearing, and contestable rather than by distributing it. Three things scope the limitation honestly. It is a property of the transition modes, not of the architecture: operating modes are country-configured states, and the designed trajectory is open, socially constructed agenda-setting. It is narrower than the passive share: engaged citizens allocate manually, delegate, or adopt configurable profiles, so authority weights fully govern only the never-engaged share. And it is now measured rather than assumed: E4 shows open construction of the weights from aggregated citizen signals is viable and scale-robust in the model, so the constraint is no longer whether distributed construction can work in principle but whether an elicitation mechanism can keep dispersed signals honest, unbiased, and representative under gaming, clientelism, and expressive-allocation pressure — a design problem the corpus gates rather than assumes away. The comparative baseline also belongs in this paragraph: under the current institutional model, the entire budget follows a centrally constructed vector that is neither published, versioned, pluggable, nor overridable by any citizen. The transition modes reproduce that centralization visibly and revocably; the open mode is designed to end it. This remains the architecture's principal open problem, now with a measured prize attached to solving it — and for that reason we treat it not as one limitation among many but as the research program's next object:

the design of open agenda-setting, including the candidate architecture in which a distributed agenda is constructed in parallel to the authority's own and the tutored role narrows to admissibility review of conflicts between the two, is the natural subject of a dedicated follow-up study and pilot.

Procedural legitimacy is not democratic mandate — and the enabling norm does not yet exist. The platform records the external authorization for budget migration and allocation formulas (the Allocation Mandate); it cannot manufacture authorization the law never granted. In the continental tradition of the reference jurisdictions, binding citizen allocation requires an enabling instrument of sufficient rank that no current statute supplies — the regional precedents (Peru's participatory-budgeting statute, Brazil's city-statute framework) prove the instrument is achievable, not that it exists — so the architecture's lawful deployments today are consultative and tutored, in which every material allocation decision remains imputed to the competent authority as a reasoned public resolution; the delivery, metering, and reputational-memory results operate unchanged under that status, and only the mature open mode requires binding allocation. The normative debate over substituting atomized allocation for representative appropriation (Rosanvallon 2008; Urbinati 2014) remains open. Under deep evaluative disagreement, the architecture's posture is procedural in Gaus's (2011) sense: its rules aim to be justifiable from diverse moral standpoints — which is what mandate records, motive neutrality, and the comparative-critique discipline provide — without presupposing a shared comprehensive doctrine. One further objection is deliberately out of scope: the model takes the coercively raised budget as given and improves its administration; the libertarian challenge to the taking itself (Nozick 1974) is neither answered nor begged here. Contribution-weighted allocation formulas, in particular, are flagged by the architecture as plutocratic departures requiring explicit higher authorization.

Fiscal dependence is measurable, not enforceable. The state controls the budget spigot. The Fiscal Commitment Profile converts strangulation from invisible to attributable — delivery latency, unexecuted valid orders, mid-cycle share cuts all become public data — but no software compels a sovereign to pay (Kydland and Prescott 1977; North and Weingast 1989). Credible commitment must come from country-level law.

Verification quality is assumed, then priced. Propositions 1–4 take detection and discovery probabilities as parameters. In thin control markets — credence-good markets where quality is unobservable to the buyer (Akerlof 1970; Dulleck and Kerschbamer 2006) — both collapse simultaneously, and the only compensating margins are financial terms and imported (remote or cross-territory) verification. The architecture prices weak verification; it cannot conjure verifiers.

Behavioral realism cuts both ways. The simulation vindicates designing for inattentive citizens, but it equally shows that a defaults-weak deployment degenerates into salience-driven allocation. Off-platform phenomena — clientelist brokerage, expressive polarization, collusion conducted entirely outside the system — are made harder and more discoverable, never impossible; the architecture's claims are comparative (against opaque monopoly), not absolute.

Meta-governance in open mode is deferred by design. Rule-change procedure, versioning, and non-surprise constraints are specified; the constitutional mechanics — rules for making rules (Buchanan and Tullock 1962) — of who votes on protocol changes in a mature open-mode deployment are deliberately not. Open-mode deployment is gated on resolving them.

Adoption selects, and the thesis does not depend on it. This paper answers whether the architecture can be built and whether it delivers more value — not whether any authority wants it. The corpus supplies the deployment layer for an authority that has decided (prospective baselines measured from instrumentation onset, credit attribution on verified delivery, institutional rather than personal timeout attribution in the first cycle, and a symmetry clause barring any operator from exempting its own projects), and names the plausible adopter archetypes — the post-scandal challenger, the mandating higher government, the conditioned external funder. The honest selection effect stands: the architecture will plausibly be adopted first by relatively clean or newly arrived sponsors, in the places that need it least.

Epistemically, this is one team's self-critiqued design. The adversarial corpus was produced by the same research effort it attacks, with AI assistance; the simulation's status-quo baseline is now audit-anchored (its parameters justified item-by-item from published audit-institution findings) but not calibrated to a specific PB dataset, and the remaining parameters are plausible rather than measured; and no pilot has been run. The three missing validations — independent expert attack, calibration to empirical PB data, and a bounded tutored pilot (sports-sector, one municipality) — are the research program's next phase, in that order.

9. Implementation pathway

The architecture is built for gradual, revocable adoption, and this section is explicit about what it does and does not claim: the research question of this paper — can the two-hundred-year-old allocation architecture be improved with today's technology, and by how much — is answered independently of whether any authority chooses to deploy; what follows is the pathway for one that has. A country opens one public function (the reference pilot is municipal sports infrastructure), migrates a small budget share under a tutored operating mode, and retains admissibility review — with every tutored decision and delay public by construction. The pilot's default framing is prospective: instrumentation begins at adoption, the visibility gap is published as the adopter's declared starting line ("measure me from here"), and pre-adoption figures are reported separately, impersonally, and as context — the configuration under which exposing instruments have historically been adopted. Functional maturity metrics (participation mix, default-flow share, fiscalization independence rates, incumbent- resistance indicators, fiscal reliability) determine whether the deployment earns wider scope, and their trajectories, not rhetoric, answer whether distribution outperforms the local baseline. The exit condition is honest in both directions: a pilot whose indicators stagnate under incumbent throttling documents that fact publicly, which is itself information the current system never yields.

The transition between regimes is itself measured. The companion experiments (Offermann 2026b) quantify the semi-open regime of the operating-regime ladder (docs/110) — a bounded mandated envelope running on protocol autopilot beside the authority's traditional budget — as a fiscal blend: above a portfolio-granularity floor of roughly ten percent, blended verified value rises monotonically and near-linearly with the envelope share, from break-even near eight to ten percent through 1.5× at half the budget to the full architecture's 2.2× at one hundred percent. The transition from the status quo toward the open regime is a dial, not a leap: adoption can proceed in increments, and every increment pays.

The same experiments measured a variable this corpus had left unregulated: *when* the authority releases budget into the allocation machinery. The resulting deployment rule: meter release against a work-in-progress ceiling calibrated to the delivery-and-verification pipeline's throughput and cycle time — never against the calendar. Calendar release freezes months of budget in escrow and saturates verification; and when verification capacity is scarce, no release policy compensates — verification capacity is the pipeline's ceiling before it is the anti-fraud instrument. The rule is conditional on a carryforward instrument (the semi-open envelope is precisely such a vehicle); under strict budget annuality it degenerates to within-year metering.

Finally, the technological premise that lowers participation costs on the citizen side (the AI tutor) applies symmetrically to the control side. Machine verification of protocolizable evidence classes multiplies fiscalization capacity, with humans as the permanent second instance — sampled re-verification with a published floor, seeded known-answer controls as the calibration and drift-detection instrument, auditing the verifier rather than competing with it — so that the machine's error rate remains measured and the human control profession remains funded from the control budget it relieves. Measured on a five-family panel of real models (Offermann 2026b), frontier models converge on good specificity and fraud detection on document-legible evidence while small local models are weaker, and evidence contracts that include objective comparison references (market benchmarks, duration bands, thresholds) let a strict verifier judge rather than guess. The machine layer reaches only document-legible, delivery-phase fraud — physical quality-below-spec and pre-contract theft remain fully human, so provenance attestation is tamper-evidence at capture, not court-grade proof, and admissibility still needs custody, contradiction, and expert testimony. Contraposed citizen evidence — independent producers with interests opposed to the executor, whose anticipated existence deters diversion — keeps the watching distributed even as routine document-verification labor shrinks; but its strength equals the *independence* of the contributor layer, and a colluding ring that captures or silences it erases the effect. Cross-layer collusion is in fact the one adversary that bypasses the per-milestone deterrence and moves leakage by an order of magnitude (while the delivered-value advantage survives), so collusion resistance — verified beneficial ownership, contributor Sybil-resistance, and decentralization of the assigner and the audit-budget floor — is a first-class requirement (`docs/113`), not a residual caution.

10. Conclusion

States do not collect taxes in order to allocate budgets; they allocate budgets in order to improve the lives of the society that funded them (Musgrave 1959). If, after a redistribution, society is no better off, the redistribution accomplished nothing — Okun's (1975) leaky bucket carried water that never arrived. The right criterion for any allocation institution is therefore not how faithfully it executes a plan but how much delivered, verified value it produces per unit of public resource. This paper's central experiments apply that criterion end to end, and their structure can be stated as two questions anyone can ask. First: take the same projects, designed identically, and change only who executes and how they are watched — does the visibly audited regime with reputational consequences deliver more than the one without them? It does, in the model: +43% delivered value on identical portfolios, because under milestone-gated verification the incentive condition holds and diversion is deterred before it happens. Second: hold the control layer fixed and change only which projects get funded, centrally planned or socially prioritized? Social prioritization delivers more under either control regime (+53-54%). The two effects multiply rather than add: verified delivery amplifies good selection, because a well-chosen project that leaks loses its advantage. And the comparison no longer rests on a caricature: recalibrated against a status-quo baseline built exclusively from the published findings of supreme audit institutions — nine jurisdictions and the European Union, administrations of every political color — the full architecture delivers, in the model, 2.22 times the calibrated status quo per unit of budget at scale, 1.4-1.6 times at municipal pilot scale, robustly across participation rates from the participatory-budgeting floor to voting-scale engagement. The recalibration produced the sharpest sentence in the paper: audit at its real, documented intensity — detection without persistent consequences — deters no diversion in the model; it shrinks the reported gap, from twenty-nine points to nineteen, never the real one. The instrument that moves delivered value is the one the status quo lacks at any audit intensity: consequences that persist. Accountability without memory is bookkeeping.

The deeper point is Friedman's: a central administration spends other people's money on other people, the spending category with the least care for both cost and value (Friedman and Friedman 1980). This architecture does not answer that problem with exhortation; it re-plumbs the bucket. Planning remains — as the guiding thread that sets scopes, floors, and mandates — but the engine of value is the conversion layer: measurable promises, conditional release, independent verification, consequences that compound into reputation, and a meter on every leak. The question this paper answers is therefore not whether states should be bigger or smaller, but whether the layers of state activity that fail through information and incentive monopoly can be re-architected to fail less — and to show their failures when they do. For one such layer we have specified a complete architecture, proved the incentive conditions its mechanisms depend on, measured selection, aggregation, and delivery in simulation against a baseline the incumbent system's own auditors supplied, and subjected the whole to four rounds of documented adversarial review with an explicit integrate-or-bound discipline. The result is deliberately modest in its claims and unusually explicit about their edges: allocation quality rides on the informational quality of whatever constructs the weight vector, whose open construction is measured viable but whose honest elicitation remains the open problem; delivered value rides on verification whose market conditions must be priced; legitimacy rides on mandates the platform can record but not create. What distinguishes the proposal is that these edges are specified, monitored, and attached to named objects — which is, we argue, what it looks like when institutional design is treated as an engineering discipline rather than an ideological one.

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Companion materials: formal proofs (`research/formal-models.md`), simulation code and full result tables (`scripts/simulation/allocation-sim.mjs`, `research/simulation-results.md`), the audit-institution evidence base (`research/audit-evidence-base.md`), the architecture corpus (`docs/`, `knowledge/`), and the complete adversarial record (`attacks/`, `defenses/`, accepted resolutions `docs/67`–`docs/109`; all forty attacks resolved and propagated).